Lecture 27

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1 The Theorem

Theorem (Cantor-Schröder-Bernstein): If $|A| \leq |B|$ and $|B| \leq |A|$, then $|A| = |B|$. In other words, if there exist injections $f : A \to B$ and $g : B \to A$, then there exists a bijection $h : A \to B$.

2 Proof Using "Ancestry"

Let $f : A \to B$ and $g : B \to A$ be injections. We analyze the "ancestry" of any element x by tracing backwards using the inverses f^{-1} and g^{-1}. Note: Since f and g are injections, inverses are unique if they exist.

For an element $a \in A$, we trace back:

$$\cdots \to g^{-1}(f^{-1}(a)) \to f^{-1}(a) \to a \to f(a) \to g(f(a)) \to \dots$$

We classify every element $x \in A$ based on where its backward chain stops:

1. **Case 1 (A-stopper):** The chain goes back to an element in A that is not in the image of g. (The "ultimate ancestor" is in A).

2. **Case 2 (B-stopper):** The chain goes back to an element in B that is not in the image of f. (The chain traces back to A, then to B, and stops there).

3. **Case 3 (Infinite/Cycle):** The chain never stops (goes back infinitely or loops).

Construction of the Bijection $F : A \to B$: We define $F(x)$ based on the classification of x:

$$F(x) = \begin{cases} f(x) & \text{if } x \text{ is an A-stopper or Infinite/Cycle} \\ g^{-1}(x) & \text{if } x \text{ is a B-stopper} \end{cases}$$

It can be proven that this piecewise function is both injective and surjective, providing the required bijection.

3 Applications of CSB

3.1 1. Equivalence of $(0, 1)$ and $(0, 1]$

We want to show $|(0, 1)| = |(0, 1]|$.

- **Injection $f : (0, 1) \to (0, 1]$:** $f(x) = x$ (Identity is an injection).
- **Injection $g : (0, 1] \to (0, 1)$:** $g(x) = x/2$ (Maps $(0, 1]$ to $(0, 0.5] \subset (0, 1)$).

By CSB, since injections exist in both directions, a bijection exists.

3.2 2. Hilbert's Hotel / Infinite Sets

If A is infinite and $x \notin A$, then $|A \cup \{x\}| = |A|$. We can map elements by shifting: $a_1 \to a_2, a_2 \to a_3 \ldots$ and placing x at a_1. This is essentially finding an injection into a proper subset.

3.3 3. Cardinality of the Power Set of Natural Numbers

Theorem: $|\mathcal{P}(\mathbb{N})| = c = |[0, 1]|$.

Proof: We construct injections in both directions.

1. **Injection $f : \mathcal{P}(\mathbb{N}) \to [0,1]$:** For a subset $A \subseteq \mathbb{N}$, define a number using decimal expansion:

$$f(A) = 0.d_1 d_2 d_3 \ldots \quad \text{where } d_i = 1 \text{ if } i \in A, \text{ else } 0$$

   (e.g., $A = \{1,3\} \to 0.10100\ldots$). This maps sets to unique real numbers (ignoring the $.100\ldots$ vs $.099\ldots$ issue for now as we only use 0s and 1s, making it strictly injective into reals).

2. **Injection $g : [0,1] \to \mathcal{P}(\mathbb{N})$:** Represent $x \in [0,1]$ in binary. This maps digits to a subset of indices.

By CSB, $|\mathcal{P}(\mathbb{N})| = |[0,1]| = c$. Since $|\mathbb{N}| = \aleph_0$, we have $\aleph_0 < c$.

4 The Continuum Hypothesis (CH)

Question: Is there a cardinality strictly between countable and continuum? Does there exist a set S such that $\aleph_0 < |S| < c$?

Hypothesis: $\aleph_1 = c$. (There is no such set).

History:

- **Cantor:** Hypothesized it is true.
- **Gödel (1931):** Proved that CH cannot be **disproved** from the standard Zermelo-Fraenkel (ZF) axioms.
- **Cohen (1963):** Proved that CH cannot be **proven** from ZF axioms.

Conclusion: CH is independent of the standard axioms of set theory.